Problem Set 2 - IO

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Question 1

Exploring the Data

```r
# Load necessary libraries
library(dplyr)
library(gtsummary)
library(ggplot2)
library(readr)
library(lubridate)
library(here)
library(kableExtra)
library(modelsummary)
library(gmm)
library(tidyr)
# Load the dataset
data <- read_csv(here("Y2S1/IO/ps2/data", "prod_data_all.csv"))
# Clean dataset to harmonize year discrepancy - 1981 = 81
data <- data %>%
  mutate(year = ifelse(year < 100, year + 1900, year)) %>%
  drop_na()

# Display the first few rows of the dataset
head(data)
```

```
# A tibble: 6 x 9
     id  year     Y    Y2     L      K     M industry country
  <dbl> <dbl> <dbl> <dbl> <dbl>  <dbl> <dbl> <chr>    <chr>
1 10250  1981  296.  113.  2.86   15.5  183. metal    col
2 10732  1989 4733. 2566. 86.2   470.  2167. metal    col
3 10732  1990 4800. 2608. 73.6   644.  2192. metal    col
```

```
4 10962  1982 7090. 1836. 37.9  5538.  5254. metal    col
5 10962  1983 4942. 1343. 40.5  6314.  3599. metal    col
6 11192  1981 1813. 1311. 24.0  1723.   502. metal    col
```

1.1

```r
# Summary statistics for each country - display no. obs for each sector and each sector over

data_col <- data %>% filter(country == "col")
data_chi <- data %>% filter(country == "chile")

# Number of observations by sector for each country
obs_by_sector_col <- data_col %>%
  group_by(industry) %>%
  summarise(n_obs = n()) %>%
  arrange(desc(n_obs))
obs_by_sector_chi <- data_chi %>%
  group_by(industry) %>%
  summarise(n_obs = n()) %>%
  arrange(desc(n_obs))
print("Colombia Observations by Sector:")
```

```
[1] "Colombia Observations by Sector:"
```

```r
print(obs_by_sector_col)
```

```
# A tibble: 5 x 2
  industry n_obs
  <chr>    <int>
1 food      6140
2 apparel   4454
3 metal     3678
4 text      2847
5 wood       964
```

```r
print("Chile Observations by Sector:")
```

```
[1] "Chile Observations by Sector:"
```

```r
print(obs_by_sector_chi)
```

```
# A tibble: 5 x 2
  industry n_obs
  <chr>    <int>
1 food     18833
2 metal     5856
3 text      5447
4 wood      4835
5 apparel   4694
```

```r
# Number of observations by year for each sector in each country
obs_by_year_sector_col <- data_col %>%
  group_by(year, industry) %>%
  summarise(n_obs = n()) %>%
  arrange(year, industry)
obs_by_year_sector_chi <- data_chi %>%
  group_by(year, industry) %>%
  summarise(n_obs = n()) %>%
  arrange(year, industry)
print("Colombia Observations by Year and Sector:")
```

```
[1] "Colombia Observations by Year and Sector:"
```

```r
print(obs_by_year_sector_col)
```

```
# A tibble: 55 x 3
# Groups:   year [11]
    year industry n_obs
   <dbl> <chr>    <int>
 1  1981 apparel    694
 2  1981 food       872
 3  1981 metal      558
 4  1981 text       429
 5  1981 wood       160
 6  1982 apparel    619
 7  1982 food       803
 8  1982 metal      514
 9  1982 text       364
10  1982 wood       146
# i 45 more rows
```

Chile: 1. Food - 2584, 2. Metal - 1038, 3. Wood - 867, 4. Apparel - 856, 5. Textile - 836
Colombia: 1. Food - 908, 2. Apparel - 744, 3. Metal - 632, 4. Textile - 475, 5. Wood - 173

Apparel Over Time - 1979: 354 → 1996: 292 - Peak at 1981 - 979

Food Over Time - 1979: 1171 → 1996: 1178 - Peak at 1981 - 1951

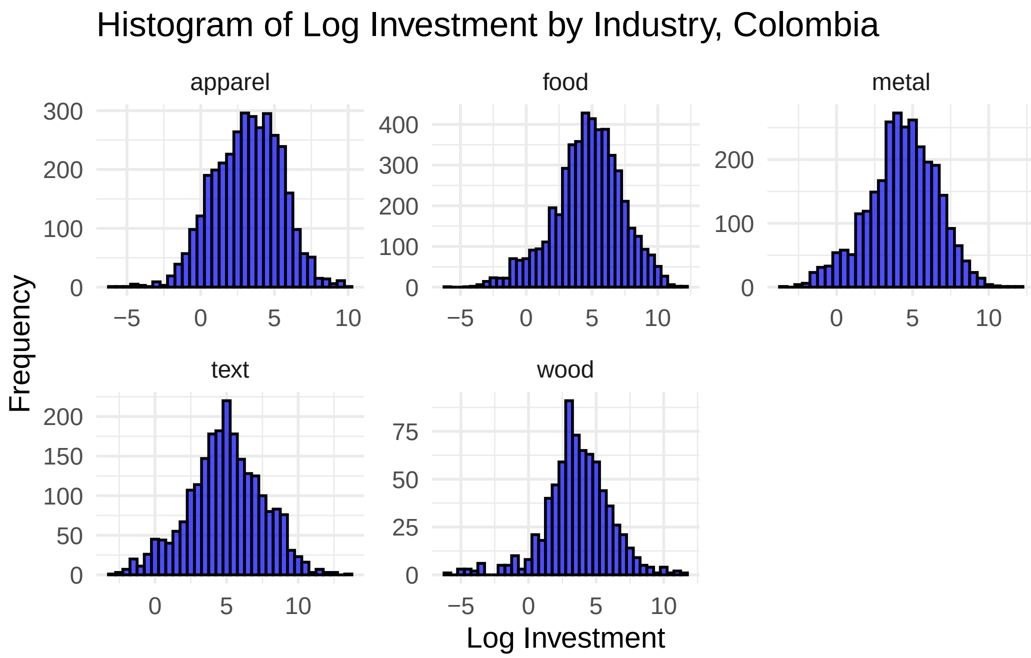
Metal Over Time - 1979: 373 → 1996: 443 - Peak at 1981 - 890

Textiles Over Time - 1979: 391 → 1996: 304 - Peak at 1981 - 752

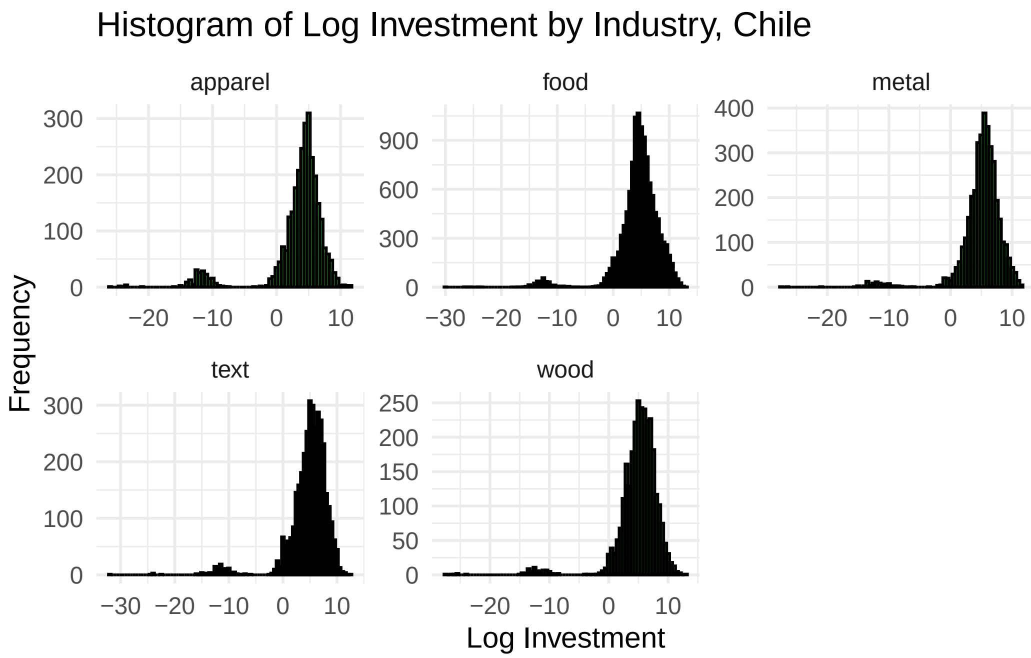
Wood Over Time - 1979: 340 → 1996: 290 - Peak at 1981 - 465

1.2

```r
# Creation of Investment Variable with 10 per depreciation rate
data <- data %>%
  arrange(id, year) %>%
  group_by(id) %>%
  mutate(
    dep    = 0.10,
    k_next = lead(K),
    i_raw  = k_next - (1 - dep) * K,
    log_i  = ifelse(i_raw  > 0, log(i_raw), NA_real_),
    log_L  = ifelse(L      > 0, log(L),     NA_real_),
    log_K  = ifelse(K      > 0, log(K),     NA_real_),
    log_Y2 = ifelse(Y2     > 0, log(Y2),    NA_real_)
  ) %>%
  ungroup()

data_col <- data %>% filter(country == "col")
data_chi <- data %>% filter(country == "chile")

# Visualize with Histogram for each industry
ggplot(data_col, aes(x = log_i)) +
  geom_histogram(binwidth = 0.5, fill = "blue", color = "black", alpha = 0.7) +
  facet_wrap(~ industry, scales = "free") +
  labs(title = "Histogram of Log Investment by Industry, Colombia",
       x = "Log Investment",
       y = "Frequency") +
  theme_minimal()
```




```r
ggsave("1.2_col.pdf", width = 10, height = 6)

ggplot(data_chi, aes(x = log_i)) +
  geom_histogram(binwidth = 0.5, fill = "green", color = "black", alpha = 0.7) +
  facet_wrap(~ industry, scales = "free") +
  labs(title = "Histogram of Log Investment by Industry, Chile",
       x = "Log Investment",
       y = "Frequency") +
  theme_minimal()
```



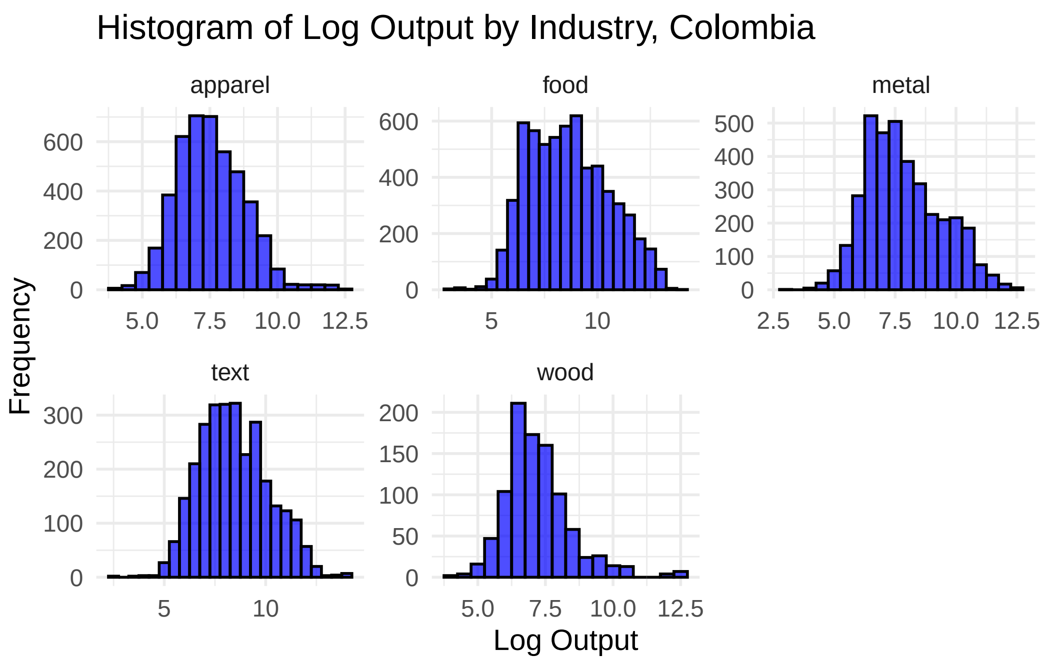

```
ggsave("1.2_chi.pdf", width = 10, height = 6)
```

In Colombia, the is much more dispersion in investment levels across industries, with some industries even experiencing negative investment. Food, metal, and textiles have very similar investment patterns, with means around 5. Apparel and wood are similar to the other three, though a bit lower. In Chile, investment levels are super consistent across industries, with all industries having means aound 5-7. Distributions are also much tighter, though most do have some negative investment observations.

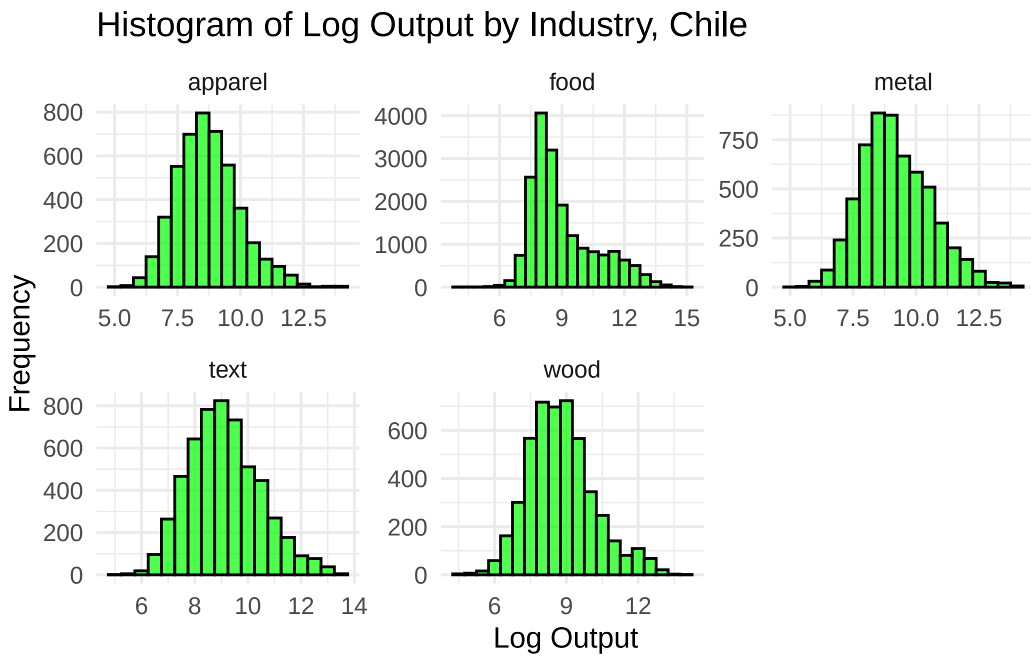
1.3

```
data_col <- data_col %>%
  group_by(industry) %>%
  mutate(
    log_Y = log(Y),
  ) %>%
  ungroup()

data_chi <- data_chi %>%
  group_by(industry) %>%
  mutate(
```

```r
    log_Y = log(Y),
  ) %>%
  ungroup()

# Visualize with Histogram for each industry
ggplot(data_col, aes(x = log_Y)) +
  geom_histogram(binwidth = 0.5, fill = "blue", color = "black", alpha = 0.7) +
  facet_wrap(~ industry, scales = "free") +
  labs(title = "Histogram of Log Output by Industry, Colombia",
       x = "Log Output",
       y = "Frequency") +
  theme_minimal()
```




```r
ggsave("1.3_col.pdf", width = 10, height = 6)

ggplot(data_chi, aes(x = log_Y)) +
  geom_histogram(binwidth = 0.5, fill = "green", color = "black", alpha = 0.7) +
  facet_wrap(~ industry, scales = "free") +
  labs(title = "Histogram of Log Output by Industry, Chile",
       x = "Log Output",
       y = "Frequency") +
  theme_minimal()
```



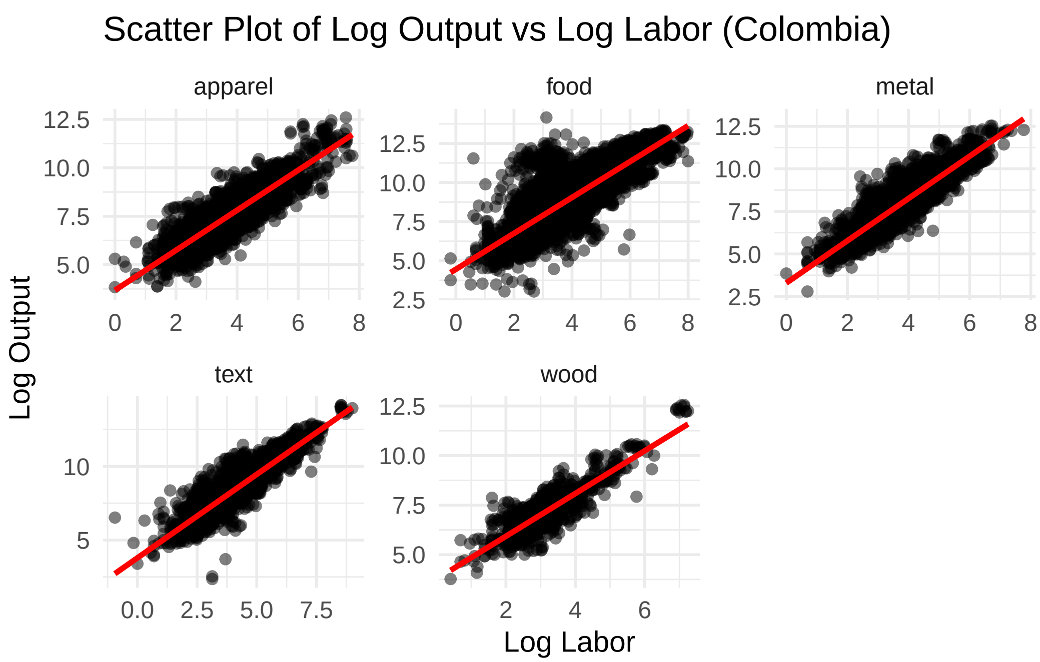

```
ggsave("1.3_chi.pdf", width = 10, height = 6)
```

In Colombia, output levels do not vary too much across industries, with means around 7-10 and similar distributions. In Chile, distribtions are also similar, though there is a bit more variation. Textiles and food have a wider distribution of output levels, while apparel and metal are a bit tighter.

1.4

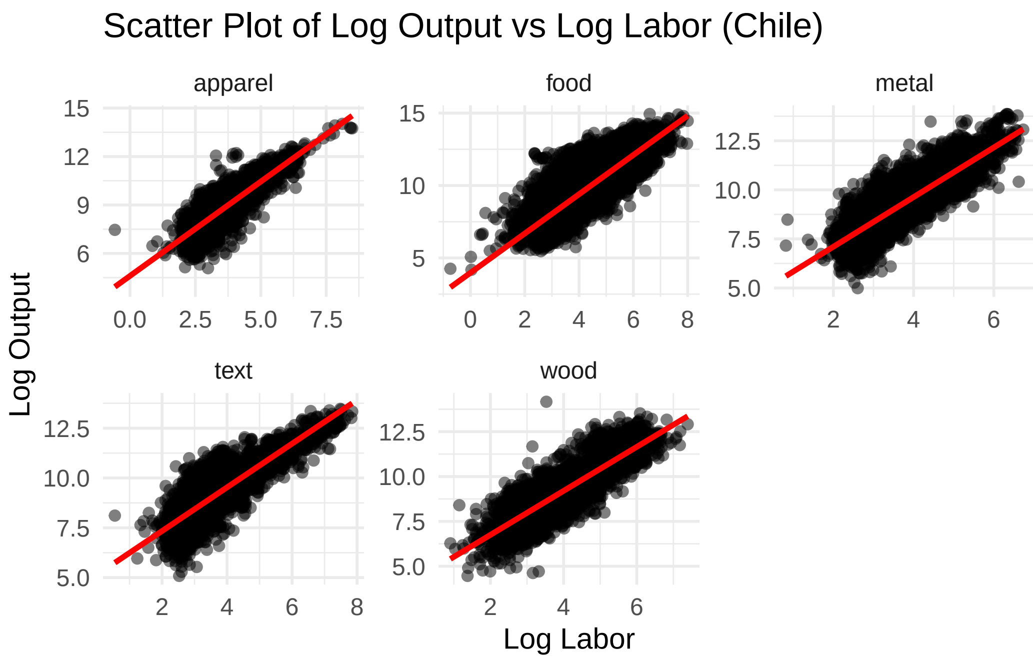
```
# Scatter plot of log_Y and log_L for each sector in each country
data_col <- data_col %>%
  group_by(industry) %>%
  mutate(
    log_L = log(L),
    log_K_ind_col = log(K),
    log_L_ind_col = log(L)
  ) %>%
  ungroup()
data_chi <- data_chi %>%
  group_by(industry) %>%
  mutate(
```

```
    log_L = log(L),
    log_K_ind_chi = log(K),
    log_L_ind_chi = log(L)
  ) %>%
  ungroup()

ggplot(data_col, aes(x = log_L, y = log_Y)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", color = "red") +
  facet_wrap(~ industry, scales = "free") +
  labs(title = "Scatter Plot of Log Output vs Log Labor (Colombia)",
       x = "Log Labor",
       y = "Log Output") +
  theme_minimal()
```




```
ggsave("1.4_col.pdf", width = 10, height = 6)

ggplot(data_chi, aes(x = log_L, y = log_Y)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", color = "red") +
  facet_wrap(~ industry, scales = "free") +
  labs(title = "Scatter Plot of Log Output vs Log Labor (Chile)",
       x = "Log Labor",
```

```
      y = "Log Output") +
  theme_minimal()
```




```
ggsave("1.4_chi.pdf", width = 10, height = 6)
```

In both countries, there is a positive relationship between labor input and output across all industries. The strength of this relationship varies by industry, with some industries showing a stronger correlation than others. For example, in Colombia, the food industry shows a strong positive correlation, while the wood industry has a weaker correlation. In Chile, all industries show a strong positive correlation.

Question 2

2.1

The estimating equation is as follows:

$$\log(Y_{it}) = \alpha_0 + \omega_{it} + \epsilon_{it} + \alpha_L \log(L_{it}) + \alpha_K \log(K_{it})$$

To get here, we take the log of:

	Colombia	Chile
(Intercept)	2.186***	2.351***
	(0.018)	(0.020)
log_L	0.831***	0.887***
	(0.006)	(0.007)
log_K	0.273***	0.303***
	(0.004)	(0.004)
Num.Obs.	18 083	39 665
R2	0.823	0.660
R2 Adj.	0.823	0.660
AIC	291 626.5	725 786.1
BIC	291 657.7	725 820.5
Log.Lik.	−18 545.704	−54 890.486
F	42 113.270	38 426.063
RMSE	0.67	0.97

* p < 0.1, ** p < 0.05, *** p < 0.01

$$Y2_{it} = e^{\alpha_0+\omega_{it}+\epsilon_{it}} L_{it}^{\alpha_L} K_{it}^{\alpha_K}$$

Where Y_{it} is the value added output of firm i at time t, L_{it} is labor input and K_{it} is capital input. The ω_{it} term is heterogeneous productivity shocks, varying across firms and years. As discussed in class, we can use OLS to estimate, but we will have some issues with endogeneity via ω_{it} showing up in the labor FOC.

```r
# Estimate homogenous parameteres using OLS for each country
ols_col <- lm(log(Y2) ~ log_L + log_K, data = data_col)

ols_chi <- lm(log(Y2) ~ log_L + log_K, data = data_chi)

modelsummary(
  list(Colombia = ols_col, Chile = ols_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)
```

```r
# Estimate varying parameters across industries using OLS for each country

ols_ind_col <- lm(log(Y2) ~ industry + log_L:industry + log_K:industry,
  data = data_col)
```

```r
ols_ind_chi <- lm(log(Y2) ~ industry + log_L:industry + log_K:industry,
  data = data_chi)

modelsummary(
  list(Colombia_Ind = ols_ind_col, Chile_Ind = ols_ind_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)
```

As can be seen in the regression outputs, when we add heterogeneity across industries, the estimates for labor and capital change slightly. However, we can now ascertain differences in productivity across industries via the industry coefficients.

2.2

```r
form_hom <- as.formula("log(Y2) ~ log_L + log(K) + log_i")

ols_inv_col <- lm(form_hom, data = data_col)
ols_inv_chi <- lm(form_hom, data = data_chi)

modelsummary(
  list(Colombia = ols_inv_col, Chile = ols_inv_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)
```

```r
form_ind <- as.formula("log(Y2) ~ industry + log_L:industry + log_K:industry + log_i:industry
ols_inv_ind_col <- lm(form_ind, data = data_col)
ols_inv_ind_chi <- lm(form_ind, data = data_chi)

modelsummary(
  list(Colombia_Ind = ols_inv_ind_col, Chile_Ind = ols_inv_ind_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)
```

When we add investment as a regressor, we see that the coefficients on labor and capital change slightly again. The investment coefficient is positive and significant across all specifications,

	Colombia_Ind	Chile_Ind
(Intercept)	2.510***	2.865***
	(0.040)	(0.063)
industryfood	−0.272***	−0.915***
	(0.049)	(0.068)
industrymetal	−0.536***	0.131
	(0.058)	(0.082)
industrytext	−0.294***	0.156*
	(0.059)	(0.083)
industrywood	0.049	−0.377***
	(0.089)	(0.085)
industryapparel × log_L	0.921***	1.055***
	(0.013)	(0.021)
industryfood × log_L	0.754***	0.896***
	(0.010)	(0.010)
industrymetal × log_L	0.998***	0.994***
	(0.016)	(0.018)
industrytext × log_L	0.858***	0.874***
	(0.015)	(0.017)
industrywood × log_L	0.974***	1.016***
	(0.030)	(0.020)
industryapparel × log_K	0.133***	0.150***
	(0.009)	(0.013)
industryfood × log_K	0.321***	0.334***
	(0.007)	(0.005)
industrymetal × log_K	0.208***	0.232***
	(0.010)	(0.009)
industrytext × log_K	0.257***	0.243***
	(0.010)	(0.010)
industrywood × log_K	0.117***	0.174***
	(0.018)	(0.010)
Num.Obs.	18 083	39 665
R2	0.831	0.689
R2 Adj.	0.831	0.689
AIC	290 826.2	722 221.0
BIC	290 951.0	722 358.4
Log.Lik.	−18 133.547	−53 095.913
F	6352.788	6275.729
RMSE	0.66	0.92

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

	Colombia	Chile
(Intercept)	2.282***	2.521***
	(0.023)	(0.026)
log_L	0.828***	0.844***
	(0.007)	(0.009)
log(K)	0.242***	0.299***
	(0.006)	(0.005)
log_i	0.031***	0.017***
	(0.004)	(0.002)
Num.Obs.	14 315	25 125
R2	0.836	0.689
R2 Adj.	0.836	0.689
AIC	232 325.5	471 395.1
BIC	232 363.3	471 435.7
Log.Lik.	−13 820.998	−33 699.484
F	24 327.769	18 593.859
RMSE	0.64	0.93

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

indicating that higher investment is associated with higher value added output. Industry level heterogeneity remains mostly significant as well.

Comparing to the results from 2.1, inclusion of investment seems to improve model fit slightly, as seen in the adjusted R-squared values, suggesting that investment is a factor worth considering when modeling firm output.

2.3

```
# First stage: estimate log_y2 = log_l + phi(log_k) s.t. phi = alpha_0 +
# alpha_k*log_k + a(i*t*k)^2 + b(i*t*k) + c

data_col <- data_col %>%
  arrange(id, year) %>%
  group_by(id) %>%
  mutate(
    log_k_next = lead(log_K),
    log_y2_next = lead(log_Y2)
  ) %>%
  ungroup()
```

	Colombia_Ind	Chile_Ind
(Intercept)	2.545***	3.106***
	(0.048)	(0.084)
industryfood	−0.161***	−1.015***
	(0.061)	(0.091)
industrymetal	−0.504***	0.101
	(0.072)	(0.109)
industrytext	−0.267***	0.253**
	(0.074)	(0.108)
industrywood	−0.027	−0.482***
	(0.114)	(0.113)
industryapparel × log_L	0.925***	1.006***
	(0.015)	(0.027)
industryfood × log_L	0.767***	0.843***
	(0.011)	(0.012)
industrymetal × log_L	0.985***	0.943***
	(0.017)	(0.022)
industrytext × log_L	0.834***	0.846***
	(0.017)	(0.020)
industrywood × log_L	0.954***	0.977***
	(0.034)	(0.025)
industryapparel × log_K	0.119***	0.146***
	(0.012)	(0.017)
industryfood × log_K	0.267***	0.339***
	(0.010)	(0.006)
industrymetal × log_K	0.196***	0.218***
	(0.014)	(0.012)
industrytext × log_K	0.230***	0.202***
	(0.014)	(0.013)
industrywood × log_K	0.139***	0.176***
	(0.028)	(0.014)
industryapparel × log_i	0.015**	0.016***
	(0.007)	(0.004)
industryfood × log_i	0.042***	0.015***
	(0.007)	(0.003)
industrymetal × log_i	0.017*	0.022***
	(0.009)	(0.005)
industrytext × log_i	0.045***	0.027***
	(0.009)	(0.004)
industrywood × log_i	−0.001	0.020***
	(0.016)	(0.005)
Num.Obs.	14 315	25 125
R2	0.843	0.715
R2 Adj.	0.843	0.715
AIC	231 710.0	469 242.0
BIC	231 869.0	469 412.8
Log.Lik.	−13 497.278	−32 606.965
F	4049.263	3320.642
RMSE	0.62	0.89

* p < 0.1, ** p < 0.05, *** p < 0.01

```r
data_chi <- data_chi %>%
  arrange(id, year) %>%
  group_by(id) %>%
  mutate(
    log_k_next = lead(log_K),
    log_y2_next = lead(log_Y2)
  ) %>%
  ungroup()

# Define a function to perform the first stage regression and get phi_hat
first_stage <- function(data) {
  data <- data %>%
    mutate(
      k_sq = log_K^2,
      it_k = as.numeric(factor(paste(id, year))) * log_K,
      it_k_sq = it_k^2,
      inv_sq = log_i^2,
      k_inv = log_K * log_i,
      k_inv_sq = k_inv^2,
      rhs = cbind(log_L, log_K, k_sq, log_i, inv_sq, k_inv, k_inv_sq)
    )
  fs_model <- lm(log_Y2 ~ log_L + rhs, data = data)
  data$bl_hat <- coef(fs_model)["log_L"]
  data$phi_hat <- predict(fs_model, newdata = data)
  return(data)
}
data_col <- first_stage(data_col)
data_chi <- first_stage(data_chi)

# Second stage: estimate y2_{t+1} - alpha_l*l_{t+1} = alpha_0 + alpha_k*k_{t+1}
# + g(phi_hat_t - alpha_k*k_t - alpha_0) + error

data_col <- data_col %>%
  mutate(
    lhs = log_y2_next - bl_hat * lead(log_L)
  )
data_chi <- data_chi %>%
  mutate(
    lhs = log_y2_next - bl_hat * lead(log_L)
  )

form_op <- as.formula("lhs ~ beta0 * betak*log_k_next + betag*I(phi_hat - betak*log_K)")
```

	Colombia	Chile
beta0	0.839***	0.755***
	(0.069)	(0.020)
betak	0.218***	0.440***
	(0.042)	(0.037)
betag	0.510***	0.522***
	(0.006)	(0.006)
Num.Obs.	14 273	25 099
AIC	31 714.9	69 831.0
BIC	31 745.2	69 863.5
Log.Lik.	−15 853.475	−34 911.508
isConv	TRUE	TRUE
finTol	5.87543307232137e-08	2.51406477231752e-09

* p < 0.1, ** p < 0.05, *** p < 0.01

```
op_col <- nls(form_op, data = data_col,
  start = list(beta0 = 0, betak = 0.3, betag = 0.3))
op_chi <- nls(form_op, data = data_chi,
  start = list(beta0 = 0, betak = 0.3, betag = 0.3))

modelsummary(
  list(Colombia = op_col, Chile = op_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)
```

```
# Estimate varying parameters across industries using OP for each country
# First stage already done above
data_col <- data_col %>%
  mutate(
    industry_factor = as.factor(industry),
    ind = as.integer(industry_factor),
    lhs = log_y2_next - bl_hat * lead(log_L)
  )
data_chi <- data_chi %>%
  mutate(
    industry_factor = as.factor(industry),
    ind = as.integer(industry_factor),
    lhs = log_y2_next - bl_hat * lead(log_L)
```

```
  )

n_ind_col <- nlevels(data_col$industry_factor)
n_ind_chi <- nlevels(data_chi$industry_factor)

form_op_ind <- as.formula("lhs ~ beta0[ind] + betak[ind]*log_k_next + betag[ind]*I(phi_hat - 
op_ind_col <- nls(form_op_ind, data = data_col,
  start = list(beta0 = rep(0, n_ind_col), betak = rep(0.3, n_ind_col), betag = rep(0.3, n_in
op_ind_chi <- nls(form_op_ind, data = data_chi,
  start = list(beta0 = rep(0, n_ind_chi), betak = rep(0.3, n_ind_chi), betag = rep(0.3, n_in

modelsummary(
  list(Colombia_Ind = op_ind_col, Chile_Ind = op_ind_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)
```

Homogenous

When comparing OP and OLS, the OP estimates for capital are lower than OLS estimates, suggesting that OLS may be overestimating the return to capital. When allowing for industry heterogeneity, capital coefficients actually increase significantly for both countries, indicating that returns to capital vary substantially across industries.

2.4

```
first_stage_lp <- function(data) {
  data <- data %>%
    mutate(
      k_sq = log_K^2,
      it_k = as.numeric(factor(paste(id, year))) * log_K,
      it_k_sq = it_k^2,
      inv_sq = log_i^2,
      k_inv = log_K * log_i,
      k_inv_sq = k_inv^2,
      log_M = log(M),
      log_M_next = lead(log_M),
      m_sq = log_M^2,
      k_m = log_K * log_M,
```

	Colombia_Ind	Chile_Ind
beta01	1.947***	2.053***
	(0.073)	(0.118)
beta02	1.944***	1.356***
	(0.046)	(0.057)
beta03	1.227***	2.480***
	(0.072)	(0.104)
beta04	1.807***	2.936***
	(0.072)	(0.089)
beta05	1.746***	1.759***
	(0.147)	(0.115)
betak1	0.093***	0.064*
	(0.015)	(0.035)
betak2	0.279***	0.357***
	(0.008)	(0.007)
betak3	0.165***	0.218***
	(0.019)	(0.019)
betak4	0.262***	0.235***
	(0.013)	(0.016)
betak5	0.058*	0.152***
	(0.035)	(0.025)
betag1	0.214***	0.337***
	(0.018)	(0.032)
betag2	0.092***	0.154***
	(0.014)	(0.015)
betag3	0.309***	0.228***
	(0.022)	(0.027)
betag4	0.122***	0.080***
	(0.021)	(0.024)
betag5	0.288***	0.263***
	(0.039)	(0.031)
Num.Obs.	14 273	25 099
AIC	28 118.9	65 007.5
BIC	28 239.9	65 137.6
Log.Lik.	−14 043.439	−32 487.742
isConv	TRUE	TRUE
finTol	6.64993877445889e-07	4.33991608884684e-06

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

```r
      k_m_sq = k_m^2,
      m_inv = log_M * log_i,
      m_inv_sq = m_inv^2,
      rhs = cbind(log_L, log_K, k_sq, log_M, m_sq, log_i, inv_sq,
                  k_inv, k_inv_sq, k_m, k_m_sq, m_inv, m_inv_sq)
    )
  fs_model <- lm(log_Y2 ~ log_L + rhs, data = data)
  data$bl_hat <- coef(fs_model)["log_L"]
  data$phi_hat <- predict(fs_model, newdata = data)
  return(data)
}

data_col_lp <- first_stage_lp(data_col)
data_chi_lp <- first_stage_lp(data_chi)

data_col_lp <- data_col_lp %>%
  mutate(
    log_y2_next = lead(log_Y2),
    log_k_next = lead(log_K),
    log_M_next = lead(log_M),
    lhs = log_y2_next - bl_hat * lead(log_L)
  )
data_chi_lp <- data_chi_lp %>%
  mutate(
    log_y2_next = lead(log_Y2),
    log_k_next = lead(log_K),
    log_M_next = lead(log_M),
    lhs = log_y2_next - bl_hat * lead(log_L)
  )

# Second stage for LP
form_lp <- as.formula("lhs ~ beta0 + betak*log_k_next + betam*log_M_next + betag*I(phi_hat -
lp_col <- nls(form_lp, data = data_col_lp,
  start = list(beta0 = 0, betak = 0.3, betam = 0.3, betag = 0.3))
lp_chi <- nls(form_lp, data = data_chi_lp,
  start = list(beta0 = 0, betak = 0.3, betam = 0.3, betag = 0.3))

modelsummary(
  list(Colombia = lp_col, Chile = lp_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)
```

	Colombia	Chile
beta0	1.131***	0.673***
	(0.023)	(0.033)
betak	0.083***	0.186***
	(0.005)	(0.006)
betam	0.350***	0.396***
	(0.005)	(0.008)
betag	0.112***	0.203***
	(0.010)	(0.018)
Num.Obs.	14 315	25 125
AIC	23 313.2	63 675.1
BIC	23 351.0	63 715.7
Log.Lik.	−11 651.584	−31 832.539
isConv	TRUE	TRUE
finTol	1.55816414427135e-06	7.12031920263528e-06

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

```
# Estimate varying parameters across industries using LP for each country
data_col_lp <- data_col_lp %>%
  mutate(
    industry_factor = as.factor(industry),
    ind = as.integer(industry_factor),
    log_M_next = lead(log_M)
  )
data_chi_lp <- data_chi_lp %>%
  mutate(
    industry_factor = as.factor(industry),
    ind = as.integer(industry_factor),
    log_M_next = lead(log_M)
  )

n_ind_col <- nlevels(data_col_lp$industry_factor)
n_ind_chi <- nlevels(data_chi_lp$industry_factor)

form_lp_ind <- as.formula("lhs ~ beta0[ind] + betak[ind]*log_k_next + betam[ind]*log_M_next +
lp_ind_col <- nls(form_lp_ind, data = data_col_lp,
  start = list(beta0 = rep(0, n_ind_col), betak = rep(0.3, n_ind_col), betam = rep(0.3, n_inc
lp_ind_chi <- nls(form_lp_ind, data = data_chi_lp,
  start = list(beta0 = rep(0, n_ind_chi), betak = rep(0.3, n_ind_chi), betam = rep(0.3, n_inc
```

```r
modelsummary(
  list(Colombia_Ind = lp_ind_col, Chile_Ind = lp_ind_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)
```

The LP estimates for capital actually come back negative for both countries, which is counterintuitive. However, intermediate inputs are positive and significant, suggesting that firms rely heavily on these inputs for production. When allowing for industry heterogeneity, capital coefficients become positive again, indicating that industry effects are important for interpretation of the results.

2.5

```r
acf_fs <- function(data) {
  data <- data %>%
    arrange(id, year) %>%
    group_by(id) %>%
    mutate(
      lag_log_L    = lag(log_L, 1),
      lag_log_K    = lag(log_K, 1),
      t2_lag_log_L = lag(log_L, 2),
      t2_lag_log_K = lag(log_K, 2)
    ) %>%
    ungroup() %>%
    mutate(
      k_sq   = log_K^2,
      l_sq   = log_L^2,
      l_k    = log_L * log_K,
      l_k_sq = l_k^2,
      log_M = log(M),
      m_sq = log_M^2,
      l_m = log_L * log_M,
      l_m_sq = l_m^2,
      k_m = log_K * log_M,
      k_m_sq = k_m^2,
      l_k_m = log_L * log_K * log_M,
      l_k_m_sq = l_k_m^2

    )
```

	Colombia_Ind	Chile_Ind
beta01	1.403***	1.153***
	(0.056)	(0.098)
beta02	0.976***	0.014
	(0.036)	(0.045)
beta03	1.040***	1.456***
	(0.054)	(0.084)
beta04	1.086***	1.536***
	(0.056)	(0.084)
beta05	1.309***	0.648***
	(0.114)	(0.094)
betak1	0.029**	0.009
	(0.012)	(0.039)
betak2	0.074***	0.155***
	(0.007)	(0.008)
betak3	0.067***	0.113***
	(0.011)	(0.020)
betak4	0.118***	0.136***
	(0.009)	(0.019)
betak5	0.047*	0.075***
	(0.025)	(0.023)
betam1	0.267***	0.221***
	(0.013)	(0.032)
betam2	0.412***	0.495***
	(0.007)	(0.010)
betam3	0.434***	0.372***
	(0.013)	(0.022)
betam4	0.451***	0.275***
	(0.011)	(0.025)
betam5	0.237***	0.441***
	(0.033)	(0.022)
betag1	0.216***	0.518***
	(0.021)	(0.037)
betag2	0.010	0.177***
	(0.016)	(0.025)
betag3	0.037	0.308***
	(0.029)	(0.043)
betag4	−0.143***	0.308***
	(0.027)	(0.037)
betag5	0.275***	0.244***
	(0.048)	(0.052)
Num.Obs.	14 315	25 125
AIC	22 553.0	60 324.9
BIC	22 712.0	60 495.7
Log.Lik.	−11 255.502	−30 141.452
isConv	TRUE	TRUE
finTol	1.58192225482312e-06	7.95736960046656e-06

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

```
  fs_model <- lm(
    log_Y2 ~ log_L + log_K + log_M + k_sq + l_sq + l_k + l_k_sq +
               m_sq + l_m + l_m_sq + k_m + k_m_sq + l_k_m + l_k_m_sq,
    data = data
  )

  data$phi_hat <- as.numeric(predict(fs_model, newdata = data))

  data
}

data_col_acf <- acf_fs(data_col)
data_chi_acf <- acf_fs(data_chi)

acf_nls <- function(df) {
  df_nls <- df %>%
    filter(
      !is.na(log_Y2),
      !is.na(log_L),
      !is.na(log_K),
      !is.na(phi_hat),
      !is.na(lag_log_K),
      !is.na(lag_log_L),
      !is.na(t2_lag_log_L)
    )

  start_vals <- list(
    beta0  = 0,
    betal  = 0.3,
    betak  = 0.3,
    betag  = 0.7
  )

  nls(
    log_Y2 ~ beta0 + betal  * log_L + betak  * log_K + betag  * (phi_hat - (betak * lag_log_
    data    = df_nls,
    start   = start_vals
  )
}

acf_col <- acf_nls(data_col_acf)
acf_chi <- acf_nls(data_chi_acf)
```

	Colombia	Chile
beta0	−0.039*	−0.162***
	(0.023)	(0.029)
betal	−0.188***	−0.087***
	(0.015)	(0.016)
betak	0.041**	0.126***
	(0.018)	(0.017)
betag	1.004***	1.018***
	(0.003)	(0.004)
Num.Obs.	12 711	28 912
AIC	20 132.3	73 546.4
BIC	20 169.5	73 587.7
Log.Lik.	−10 061.141	−36 768.190
isConv	TRUE	TRUE
finTol	1.00509932294479e-06	1.60360049606987e-07

* p < 0.1, ** p < 0.05, *** p < 0.01

```
modelsummary(
  list(Colombia = acf_col, Chile = acf_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)
```

```
# Estimate varying parameters across industries using ACF for each country
data_col_acf <- data_col_acf %>%
  mutate(
    industry_factor = as.factor(industry),
    ind = as.integer(industry_factor)
  )
data_chi_acf <- data_chi_acf %>%
  mutate(
    industry_factor = as.factor(industry),
    ind = as.integer(industry_factor)
  )
n_ind_col <- nlevels(data_col_acf$industry_factor)
n_ind_chi <- nlevels(data_chi_acf$industry_factor)

form_acf_ind <- as.formula("log_Y2 ~ beta0[ind] + betal[ind] * log_L + betak[ind] * log_K + l
acf_ind_col <- nls(form_acf_ind, data = data_col_acf,
```

```
  start = list(beta0 = rep(0, n_ind_col), betal = rep(0.3, n_ind_col), betak = rep(0.3, n_inc
acf_ind_chi <- nls(form_acf_ind, data = data_chi_acf,
  start = list(beta0 = rep(0, n_ind_chi), betal = rep(0.3, n_ind_chi), betak = rep(0.3, n_inc
modelsummary(
  list(Colombia_Ind = acf_ind_col, Chile_Ind = acf_ind_chi),
  stars = c('*' = 0.1, '**' = 0.05, '***' = 0.01),
  output = "kableExtra"
)
```

ACF estimates for capital are lower than OLS, LP, and OP estimates, suggesting that previous methods may have overestimated the return to capital. Further, labor is estimated to be lower as well, indicating that both inputs may have been overvalued. Chile also shows insignifcant estimates for capital and labor.

When allowing for industry heterogeneity, capital coefficients increase significantly for both countries, indicating that returns to capital vary substantially across industries. However, significance levels are still inconsistent, suggesting that further investigation is needed. Labor estimates are more consistent and significant across industries, also increasing from the homogenous specification.

Question 3

3.1

```
# Using OP estimates from 2.3 to estimate omega = exp(phi_it - alpha_0 - alpha_k*log(K_it))

alpha_0_col <- coef(op_col)[["beta0"]]
alpha_k_col <- coef(op_col)[["betak"]]
data_col <- data_col %>%
  mutate(
    omega_it = exp(phi_hat - alpha_0_col - alpha_k_col * log_K)
  )
alpha_0_chi <- coef(op_chi)[["beta0"]]
alpha_k_chi <- coef(op_chi)[["betak"]]
data_chi <- data_chi %>%
  mutate(
    omega_it = exp(phi_hat - alpha_0_chi - alpha_k_chi * log_K)
  )
data_col <- data_col %>%
  mutate(
```

	Colombia_Ind	Chile_Ind
beta01	0.094*	0.186**
	(0.049)	(0.086)
beta02	−0.143***	−0.585***
	(0.033)	(0.036)
beta03	−0.246***	0.664***
	(0.048)	(0.073)
beta04	−0.057	0.424***
	(0.050)	(0.070)
beta05	0.180*	−0.015
	(0.106)	(0.079)
betal1	−0.170***	−0.044
	(0.028)	(0.046)
betal2	−0.228***	−0.131***
	(0.020)	(0.018)
betal3	−0.074**	−0.054
	(0.029)	(0.041)
betal4	−0.147***	−0.093**
	(0.034)	(0.046)
betal5	0.019	−0.046
	(0.063)	(0.039)
betak1	0.021	0.138***
	(0.031)	(0.046)
betak2	0.043*	0.120***
	(0.023)	(0.019)
betak3	0.064*	0.155***
	(0.033)	(0.039)
betak4	0.018	0.015
	(0.036)	(0.038)
betak5	0.026	0.085**
	(0.071)	(0.037)
betag1	0.986***	0.989***
	(0.007)	(0.013)
betag2	0.999***	1.047***
	(0.004)	(0.005)
betag3	1.055***	0.983***
	(0.007)	(0.011)
betag4	1.020***	0.970***
	(0.006)	(0.009)
betag5	0.980***	0.967***
	(0.017)	(0.011)
Num.Obs.	15 232	33 653
AIC	23 661.1	82 656.9
BIC	23 821.4	82 833.8
Log.Lik.	−11 809.554	−41 307.456
isConv	TRUE	TRUE
finTol	4.07252025436782e-06	5.84701995106551e-06

* p < 0.1, ** p < 0.05, *** p < 0.01

```
    mean_omega_col = mean(omega_it, na.rm = TRUE),
    sd_omega_col = sd(omega_it, na.rm = TRUE)
  )
data_chi <- data_chi %>%
  mutate(
    mean_omega_chi = mean(omega_it, na.rm = TRUE),
    sd_omega_chi = sd(omega_it, na.rm = TRUE)
  )

print(paste("Colombia - Mean Omega:", data_col$mean_omega_col[1], "SD Omega:", data_col$sd_o
```

```
[1] "Colombia - Mean Omega: 294.293581267125 SD Omega: 702.900302599767"
```

```
print(paste("Chile - Mean Omega:", data_chi$mean_omega_chi[1], "SD Omega:", data_chi$sd_omeg
```

```
[1] "Chile - Mean Omega: 74.8760195478157 SD Omega: 82.6386074307829"
```

As can be seen, Colombia is more productive on average, with a mean ω of ≈ 110.33 compared to Chile's mean of ≈ 81.73. However, Colombia also has a higher standard deviation of ≈ 310.55 versus Chile's ≈ 153.75, indicating greater variability in productivity among Colombian firms.

3.2

```
## Colombia: extract industry-specific OP parameters
coef_col_ind <- coef(op_ind_col)

beta0_col_ind  <- coef_col_ind[grep("^beta0", names(coef_col_ind))]
betak_col_ind  <- coef_col_ind[grep("^betak", names(coef_col_ind))]
# betag_col_ind <- coef_col_ind[grep("^betag", names(coef_col_ind))] # not needed for omega

# Make sure 'ind' matches the order of industry_factor levels used in nls
# ind = as.integer(industry_factor) is already set upstream

data_col <- data_col %>%
  mutate(
    alpha0_ind = beta0_col_ind[ind],
    alphaK_ind = betak_col_ind[ind],
```

```r
    omega_it_het = exp(phi_hat - alpha0_ind * 1 - alphaK_ind * log_K)
  )

## Chile
coef_chi_ind <- coef(op_ind_chi)

beta0_chi_ind <- coef_chi_ind[grep("^beta0", names(coef_chi_ind))]
betak_chi_ind <- coef_chi_ind[grep("^betak", names(coef_chi_ind))]

data_chi <- data_chi %>%
  mutate(
    alpha0_ind = beta0_chi_ind[ind],
    alphaK_ind = betak_chi_ind[ind],
    omega_it_het = exp(phi_hat - alpha0_ind * 1 - alphaK_ind * log_K)
  )

omega_industry_col <- data_col %>%
  group_by(industry) %>%
  summarise(
    mean_omega = mean(omega_it_het, na.rm = TRUE),
    sd_omega   = sd(omega_it_het, na.rm = TRUE),
    .groups = "drop"
  )

omega_industry_chi <- data_chi %>%
  group_by(industry) %>%
  summarise(
    mean_omega = mean(omega_it_het, na.rm = TRUE),
    sd_omega   = sd(omega_it_het, na.rm = TRUE),
    .groups = "drop"
  )

print("Colombia - Mean and SD Omega by Industry:")
```

```
[1] "Colombia - Mean and SD Omega by Industry:"
```

```r
print(omega_industry_col)
```

```
# A tibble: 5 x 3
  industry mean_omega sd_omega
  <chr>         <dbl>    <dbl>
```

```
1 apparel       163.      376.
2 food           61.7     115.
3 metal         238.      400.
4 text          124.      279.
5 wood          294.     1310.
```

```r
print("Chile - Mean and SD Omega by Industry:")
```

```
[1] "Chile - Mean and SD Omega by Industry:"
```

```r
print(omega_industry_chi)
```

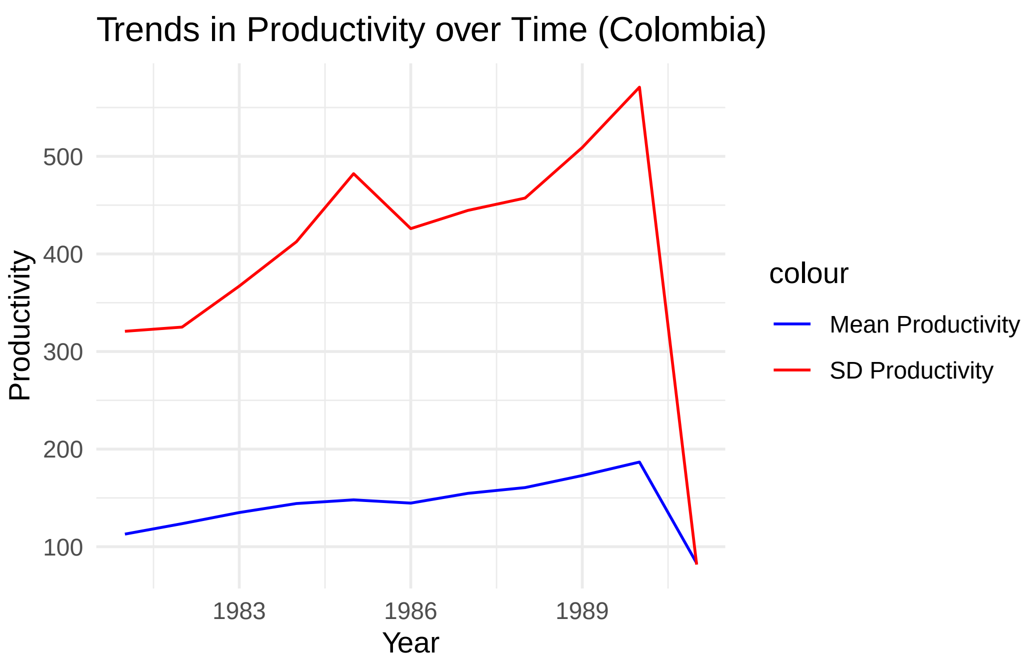
```
# A tibble: 5 x 3
  industry mean_omega sd_omega
  <chr>         <dbl>    <dbl>
1 apparel       460.     1849.
2 food           90.9     124.
3 metal          88.7     119.
4 text           62.7     134.
5 wood          357.      619.
```

Allowing for industry heterogeneity, we see that Colombia still has higher average productivity across all industries compared to Chile. In both countries, the wood industry has the highest average productivity (≈ 341 for Colombia and ≈ 54 for Chile), while the metal industry has the lowest for Colombia (≈ 100), while it is apparel for Chile (≈ 23). Standard deviations are also higher in Colombia across all industries, indicating greater variability in productivity among Colombian firms within each industry.

3.3

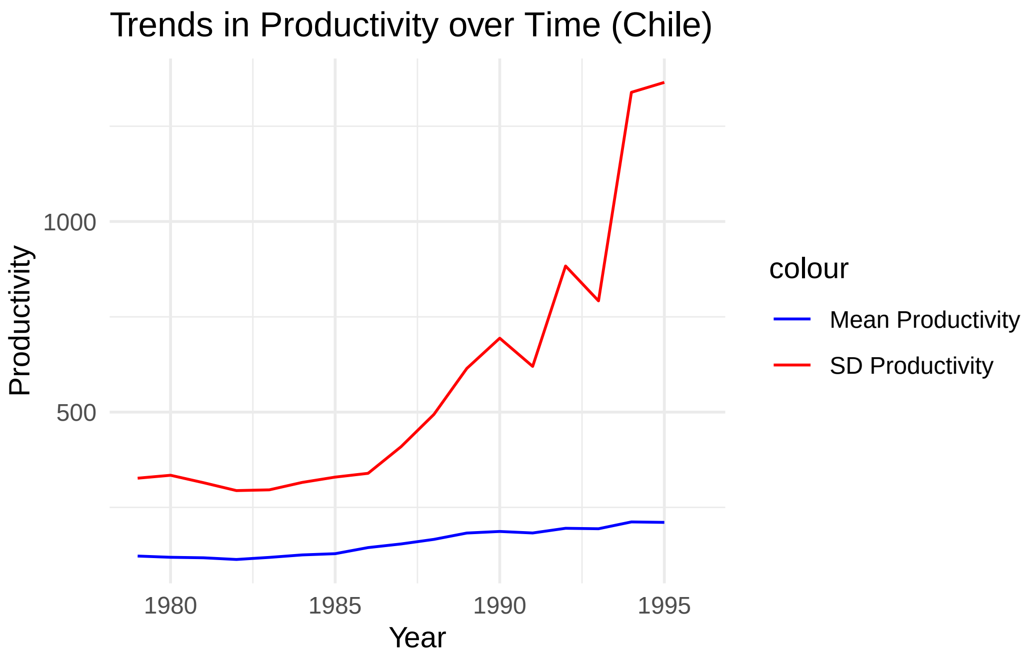
```r
# Calculate yearly mean and sd of omega_it_het for each country
prod_trends_col <- data_col %>%
  group_by(year) %>%
  summarise(
    mean_omega = mean(omega_it_het, na.rm = TRUE),
    sd_omega   = sd(omega_it_het, na.rm = TRUE),
    .groups = "drop"
  )
prod_trends_chi <- data_chi %>%
```

```
  group_by(year) %>%
  summarise(
    mean_omega = mean(omega_it_het, na.rm = TRUE),
    sd_omega   = sd(omega_it_het, na.rm = TRUE),
    .groups = "drop"
  )
# Plot trends over time for each country
ggplot(prod_trends_col, aes(x = year)) +
  geom_line(aes(y = mean_omega, color = "Mean Productivity")) +
  geom_line(aes(y = sd_omega, color = "SD Productivity")) +
  labs(title = "Trends in Productivity over Time (Colombia)",
       x = "Year",
       y = "Productivity") +
  scale_color_manual(values = c("Mean Productivity" = "blue", "SD Productivity" = "red")) +
  theme_minimal()
```



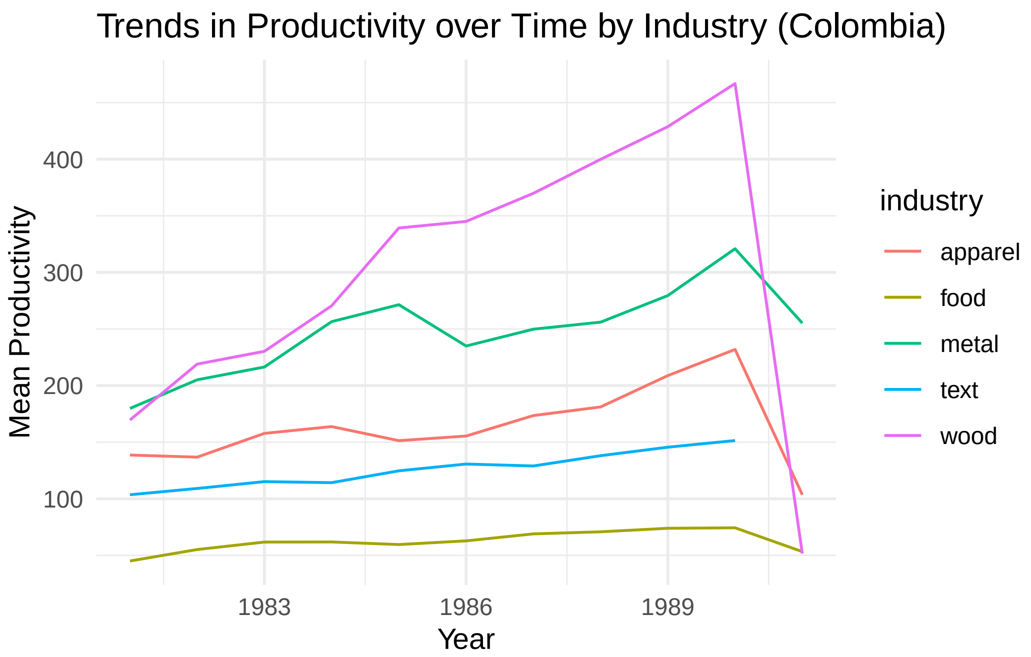

```
ggsave("3.3_hom_col.pdf", width = 10, height = 6)
ggplot(prod_trends_chi, aes(x = year)) +
  geom_line(aes(y = mean_omega, color = "Mean Productivity")) +
  geom_line(aes(y = sd_omega, color = "SD Productivity")) +
  labs(title = "Trends in Productivity over Time (Chile)",
       x = "Year",
       y = "Productivity") +
```

```r
  scale_color_manual(values = c("Mean Productivity" = "blue", "SD Productivity" = "red")) +
  theme_minimal()
```

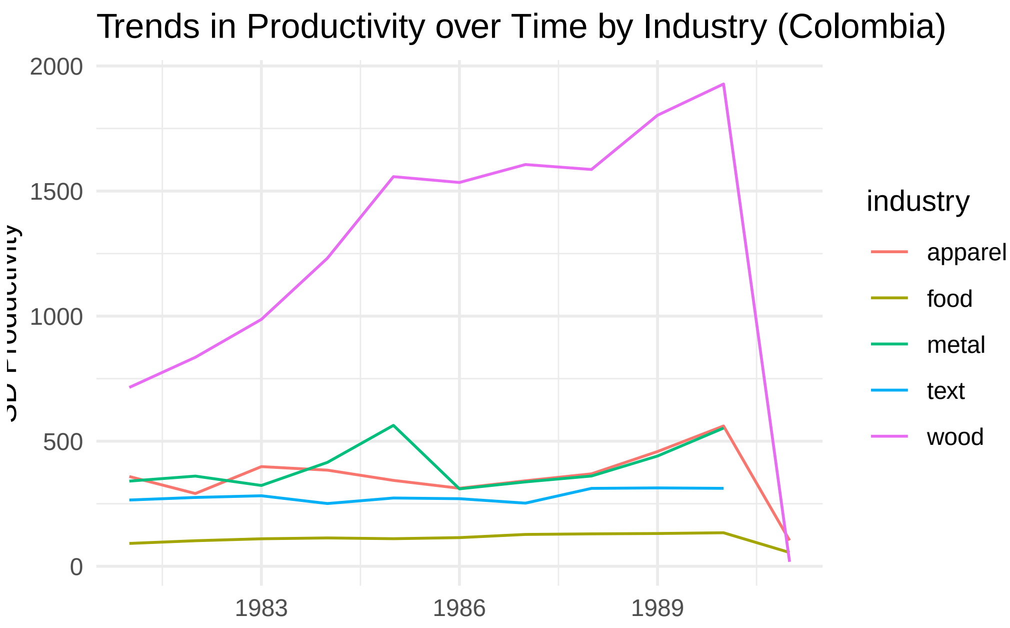



```r
ggsave("3.3_hom_chi.pdf", width = 10, height = 6)

# Now by industry for each country
prod_trends_ind_col <- data_col %>%
  group_by(year, industry) %>%
  summarise(
    mean_omega = mean(omega_it_het, na.rm = TRUE),
    sd_omega   = sd(omega_it_het, na.rm = TRUE),
    .groups = "drop"
  )
prod_trends_ind_chi <- data_chi %>%
  group_by(year, industry) %>%
  summarise(
    mean_omega = mean(omega_it_het, na.rm = TRUE),
    sd_omega   = sd(omega_it_het, na.rm = TRUE),
    .groups = "drop"
  )
# Plot trends over time by industry for each country
ggplot(prod_trends_ind_col, aes(x = year)) +
  geom_line(aes(y = mean_omega, color = industry)) +
```

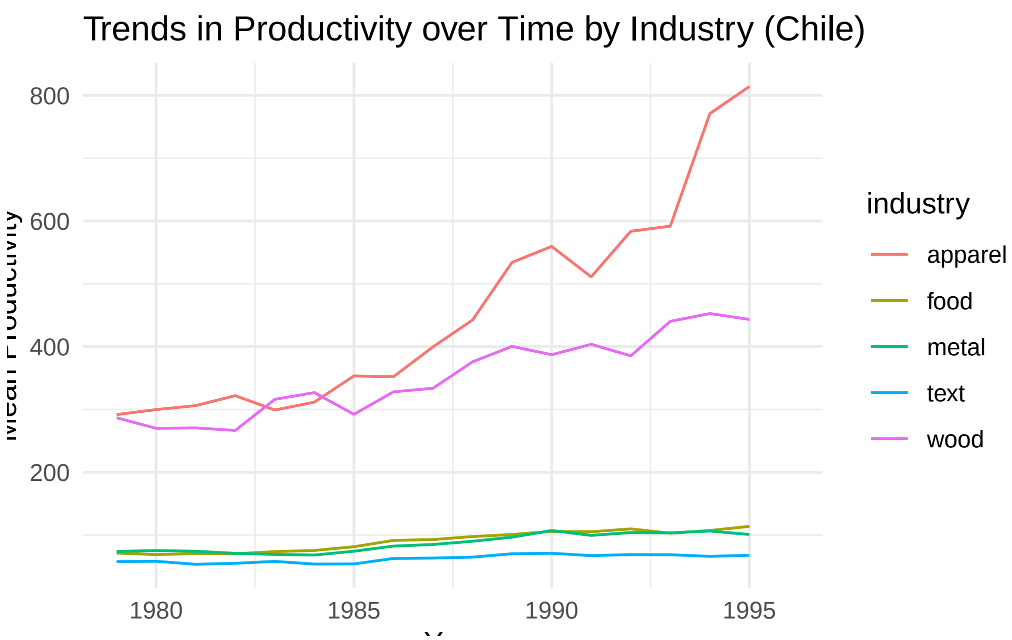
```
  labs(title = "Trends in Productivity over Time by Industry (Colombia)",
       x = "Year",
       y = "Mean Productivity") +
  theme_minimal()
```



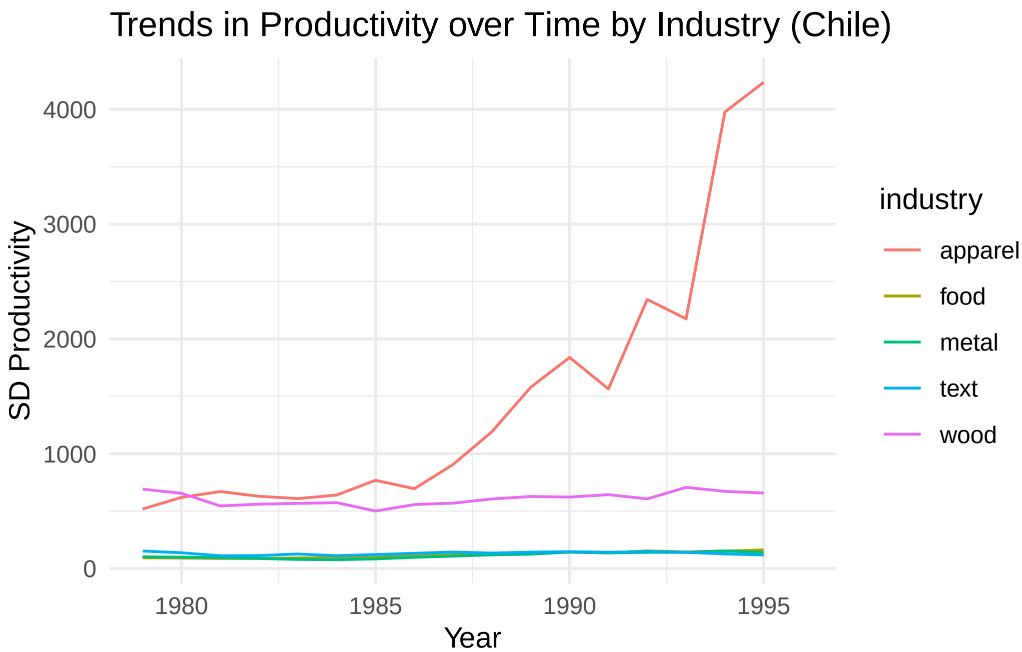

```
ggsave("3.3_ind_col_mean.pdf", width = 10, height = 6)
ggplot(prod_trends_ind_col, aes(x = year)) +
  geom_line(aes(y = sd_omega, color = industry)) +
  labs(title = "Trends in Productivity over Time by Industry (Colombia)",
       x = "Year",
       y = "SD Productivity") +
  theme_minimal()
```




```
ggsave("3.3_ind_col_sd.pdf", width = 10, height = 6)
ggplot(prod_trends_ind_chi, aes(x = year)) +
  geom_line(aes(y = mean_omega, color = industry)) +
  labs(title = "Trends in Productivity over Time by Industry (Chile)",
       x = "Year",
       y = "Mean Productivity") +
  theme_minimal()
```




```r
ggsave("3.3_ind_chi_mean.pdf", width = 10, height = 6)
ggplot(prod_trends_ind_chi, aes(x = year)) +
  geom_line(aes(y = sd_omega, color = industry)) +
  labs(title = "Trends in Productivity over Time by Industry (Chile)",
       x = "Year",
       y = "SD Productivity") +
  theme_minimal()
```




```r
ggsave("3.3_ind_chi_sd.pdf", width = 10, height = 6)
```

In Colombia, average productivity trends upward over the time horizon, with a slightly lesser increase in standard deviation, indicating that the firms are becoming more productive on average with time, while variability is not increasing to the same degree. In Chile, we see a a spike in productivity after 1985, continuing until 1990 before slowing. Standard deviation also increases over time, with more consistency in growth. This suggests that Chilean firms are also becoming more productive on average, but with greater variability among firms.

When splitting out by industry, we see that Colombian textile industries are the most productive on average, while wood and apparel are the least. However, teh textile industry also exhibits the highest variability, while food is the least so. In Chile, all industries are increasing in productivity over time to similar degrees, with textiles being the most productive until about 1992 when food overtakes it. Apparel is the least productive until about 1994, when metal takes over as least prductive. Food and apparel are the most variable industries in Chile when looking at growth in standard deviation, though textiles have the highest level until about 1990. Over the whole horizon, wood and metal are the least variable.